

# Vedant Ranganathan

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## EDUCATION

### McGill University

Montreal, QC

*Bachelor of Engineering in Mechanical Engineering, Minor in Aerospace Engineering* Aug. 2023 - May 2028 (Exp.)

## TECHNICAL EXPERIENCE

### Mechanical Engineering Intern | Manastu Space Technologies

June - Aug. 2026

- Sized an aluminum hydrogen-peroxide propellant tank and performed orbital and spacecraft thermal analysis to determine a 3.6 W worst-case heat load while maintaining the tank below 15°C.
- Designed the passive thermal-control system in **Fusion 360** using an aluminum saddle, heat pipes, and a radiator, then verified its performance through analytical calculations and **ANSYS thermal** simulations.
- Modeled a solenoid in **ANSYS Maxwell** and compared analytical and simulated magnetic-field results to develop methods applicable to Hall-effect thruster magnetic-circuit analysis.
- Designed a custom mounting structure in **SolidWorks** and integrated a tank, thruster, and valves into an electric-propulsion assembly within a constrained spacecraft envelope.

### Carbon Rims Subteam Member | McGill Formula Electric

Aug. 2025 - Present

- Supported the manufacture of carbon-fiber composite rims for a Formula Electric vehicle through mold preparation, carbon-fiber layup, trimming, finishing, and hands-on fabrication.
- Created 3D models of aerodynamic vehicle components in **Siemens NX**, supporting the design and integration of the Formula Electric car's aero package.

## LEADERSHIP & ACTIVITIES

### VP Internal | McGill Institute for Aerospace Engineering

Sep. 2025 - Present

- Coordinated an Aerospace Career Fair with 200+ attendees and 15+ industry participants including Airbus, Bombardier, Pratt & Whitney, and MDA Space.
- Led logistics and technical event operations including room scheduling, participant coordination, workshop planning, and industry-sponsored events.
- Hosted an Airbus-sponsored Airfoil Design Challenge featuring an Arduino-based aerodynamic test rig with controllable airflow and sensor instrumentation.

### Flight Student | Ponderosa Aero Club

June 2022 - Present

- Logged 30+ flight hours in a Cessna 172 and completed formal ground school covering aerodynamics, aircraft systems, meteorology, navigation, aviation regulations, and flight operations.
- Conducted full aircraft operations including preflight inspection, engine run-up, taxi, takeoff/landing, navigation, and radio communication in towered and non-towered environments, including Class C airspace at Boise Airport.
- Performed fundamental flight maneuvers including stalls/recoveries, slow flight, steep turns, slips, crosswind landings, and ground reference maneuvers.

## PROJECTS

### C-Clamp | SolidWorks, Mastercam

March 2026

- Designed and manufactured a C-clamp assembly consisting of two milled mold halves, a 3D-printed clamp body, and a turned side-action component.
- Modeled assemblies in **SolidWorks** and generated CNC milling and turning toolpaths in **Mastercam** for machining.

### Geothermal Power Plant Analysis | MATLAB, Research Paper

Jan - April 2026

- Evaluated geothermal power generation as a lower-emission alternative to a conventional methane combustion Rankine cycle power plant, estimating a potential CO<sub>2</sub> emissions reduction of approximately 75%.
- Designed a heat pump system to recover low-temperature waste heat from a geothermal power plant condenser for potential district heating and residential space heating applications.

## TECHNICAL SKILLS

**CAD/Simulation:** SolidWorks, Siemens NX, Fusion 360, AutoCAD, OnShape, ANSYS Maxwell & Mechanical

**CAM/Manufacturing:** Mastercam, G-Code, Basic CNC Operation, 3D Printing

**Programming/Other:** Python, JavaScript, MATLAB, Google Workspace, Microsoft Office